

Homework 5

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1. If G is a finitely-generated abelian group, G is isomorphic to a direct product of cyclic groups. However, the decomposition as a product of cyclic groups is not in general unique.

(a) Find an element of order 30 in the group $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$.

(b) Use your answer to (a) to prove that $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/30\mathbb{Z}$.

Proof.

(a) The element $(1, 1)$ has order 30.

(b) Since the order of $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ is 30, part (a) shows that it is cyclic since it is generated by one element. Thus $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/30\mathbb{Z}$ since cyclic groups of the same order are isomorphic.

□

2. Let C be a non-singular cubic curve given by $y^2 = f(x) = x^3 + ax^2 + bx + c$.

(a) Prove that

$$\frac{d^2y}{dx^2} = \frac{2f''(x)f(x) - f'(x)^2}{4yf(x)} = \frac{\psi_3(x)}{4yf(x)}.$$

Use this to deduce that a point $P = (x, y) \in C$ is a point of order 3 if and only if $P \neq \mathcal{O}$ and P is a point of inflection on the curve C .

(b) Suppose now that $a, b, c \in \mathbb{R}$. Prove that $\psi_3(x)$ has exactly two real roots, say α_1 and α_2 with $\alpha_1 < \alpha_2$. Show that $f(\alpha_1) < 0$ and $f(\alpha_2) > 0$. (From this it follows that $C[3](\mathbb{R}) \cong \mathbb{Z}/3\mathbb{Z}$.)

Proof.

(a) Indeed,

$$f(x) = x^3 + ax^2 + bx + c = y^2$$

$$f'(x) = 3x^2 + 2ax + b = 2y \frac{dy}{dx}$$

$$f''(x) = 6x + 2a$$

and

$$\begin{aligned}
y^2 &= x^3 + ax^2 + bx + c \\
\implies 2y \frac{dy}{dx} &= 3x^2 + 2ax + b \\
\implies \frac{dy}{dx} &= \frac{3x^2 + 2ax + b}{2y} \\
\implies \frac{d^2y}{dx^2} &= \frac{(6x + 2a)2y - 2\frac{dy}{dx}(3x^2 + 2ax + b)}{4y^2} \\
&= \frac{f''(x)2y - \frac{f'(x)}{y}f'(x)}{4f(x)} \\
&= \frac{f''(x)2y^2 - f'(x)^2}{4yf(x)} \\
&= \frac{2f''(x)f(x) - f'(x)^2}{4yf(x)} \\
&= \frac{\psi_3(x)}{4yf(x)}.
\end{aligned}$$

Certainly, P is of order 1 if and only if $P = \mathcal{O}$ since $\mathcal{O}$ is the identity, so $P \neq \mathcal{O}$ if P is of order 3. By definition, points of inflection are points where the second derivative is equal to 0, and the expression above is equal to 0 if and only if $\psi_3(x) = 0$ and $y \neq 0$ (since $y = 0$ if and only if $f(x) = 0$). Indeed, if y were 0, then P would be of order 2, not 3, so we can omit the restriction on y and determine that the points of order 3 are exactly the points of inflection which are not $\mathcal{O}$.

(b)

□

3. * Construct yourself an elliptic curve with as high rank as you can. For MTH 346 students, full credit will be awarded for a curve of rank ≥ 4 . For MTH 646 students, full credit will be awarded for a curve of rank ≥ 5 .

You should use Magma to test the rank of your elliptic curve.

When I say construct such a curve yourself, I mean you should *not* just look up curves in databases of curves with high rank. You should use some method to search. Just testing all curves in a box (say those of the form $y^2 = x^3 + Ax + B$ where $A, B \in \mathbb{Z}$ and $0 \leq A, B \leq 100$) will likely work for finding a rank 4 curve. For rank ≥ 5 you need a way to decrease the number of curves you are testing, or find a systematic way to construct curves with lots of rational points on them.